Appendix A: Supplementary Model Information

1. Summary of different ontology-driven smart systems (Table A1)

Table A 1. Review on ontology-driven smart systems (Integration of ontology and AI techniques)

Sr. No.	Model	Domain	Purpose	Significance	Approach	Citations
1.	A semantic framework consisting of "RAInS" and "SAO" ontologies is developed for AI systems.	AI	AI system design and implementation	Provide a visual interface for designing accountability plans and managing accountability records.	Modeled the accountability information relevant to the design stage of the AI system.	[72]
2.	A deep learning-based NLP ontology population system is developed.	Biology	Systems and processes analysis	Automatic population of the Biomolecular Network Ontology (BNO).	Used the popular deep learning algorithm "Word2vec" to learn word embeddings using a shallow neural network	[73]
2	An ontology model "OntoSenticNet" is proposed consisting of a composite deep learning classifier.	Text analysis	Sentiment mining	Deep learning-based text mining with ontology-based information.	Used semantic information in ontologies for explaining deep text mining, using neural attention and word embeddings.	[74]
3	A lightweight dynamic ontology "LiO-IoT" using machine learning technique is proposed for IoT systems.	IoT	Fulfilling the requirements of IoT systems.	Ensure semantic interoperability among IoT devices and reducing complexity and introducing dynamicity in the ontology.	Used machine learning technique (clustering) to provide dynamic semantics automatically for including additional concepts.	[75]
4	A novel ontology model "SOHO" is proposed for human-robot collaboration.	Robotics	A flexible representation of collaborative production processes	Enhance the flexibility of control solutions of robot-based systems.	Employed context-based approach for ontology design.	[76]
5	A smart ontology-based IoT framework is proposed for the early detection of COVID-19 in patients.	Healthcare	Covid-19 detection and remote patient monitoring.	Control the spread of the coronavirus.	An ontology-based biosensor is developed using sensory 1D biomedical Signals (ECG, PPG, temperature, and accelerometer) and machine learning techniques such as feature extraction and classification.	[34]
6	A novel approach is proposed for generating ontology using a deep learning model.	Cyber security	Ontology learning and application for cyber-attack detection.	Improve intelligent intrusion detection for cybersecurity.	Used deep neural network and supervised classifier gradient boosting decision tree (GB) for ontology learning and application.	[77]
7	A decision support system is developed using ontology and machine learning techniques to predict disease and suggest solutions.	Healthcare	Prediction of diabetes stage in patients	Improve the healthcare diagnostic system.	Used ontology containing data such as disease symptoms, causes and treatments, Naive Bayes algorithm, and decision tree for identifying disease stage.	[78]
8	An ontology model is proposed for cyber-security of self-recovering smart grids.	Smart Grids	Monitoring and restoration of the operability of the Smart Grid power systems.	Ensure sustainability and functioning of power systems in the event of destabilization	Used conceptual bases of self-recovery of perspective energy systems in the conditions of information confrontation are employed to develop an ontology model.	[33]
9	An ontology model is proposed to represent Digital Twin (DT) in the context of a cyber-physical system (CPS).	Cyber-physical system	Analysis of DT through knowledge representation.	Highlights the main concepts involved in the development of DT.	Used concepts already existing in the literature, such as the IoT-Lite ontology.	[79]

10	A smart fisheries ontology is developed, and an Attribute-Based Access Control System (ABAC) is implemented to evaluate access requests in farms.	Aquaculture	Protect the internet-connected sensors from potential cyber-attacks and propose AI applications to aid the owners to effectively manage their fish farms.	Secure and efficient smart fisheries ecosystem.	Used an ecosystem consisting of virtual and digital entities and their interactions to develop an ontology model, which is then used to implement ABAC.	[46]
11	A method based on natural language analysis and big data ontology is proposed for metadata analysis,	Big data	Analysis of metadata records in big data.	Improve the efficiency of all stages of big data processing	Used data created by human experts and generated by electronic devices to populate the instances of the ontological big data model.	[80]
12	A decision support system based on ontology-based text mining, self-organizing maps (machine learning algorithm), reliability and cost optimization is developed for design improvement using warranty data.	Design	Identification of manufacturing faults and optimization of design parameters	Enhanced design reliability and reduced manufacturing costs.	Used ontology-based text mining approach to extract hidden knowledge in warranty database, data mining approach SOM to link information with manufacturing data, and statistical analysis for re-evaluation of parameters for cost analysis and design changes.	[81]
13	A system “PALS (Privacy via Anomaly-detection System)” is developed based on ABAC and ontology model to execute access control decisions for smart homes.	Smart home	Detect the intrusions in the smart home.	Enable users to control devices at home by providing them with the ability to change the access control policy.	Used context-sensitive policies that were created based on online privacy policies available from cloud service providers like Google Nest suite of products.	[36]
14	Innovative ontology matching system is developed that finds complex correspondences by processing expert knowledge from external domain ontologies and by using novel matching methods.	Smart grids	Identify and solve interoperability issues within smart grids.	Improve performance of smart grids.	Used innovative graph-based matching methods and innovative mapping algorithms.	[82]
15	A novel ontology-based neural network model “OntoLSTM” is proposed for manufacturing time series classification.	Smart factory	Enhancing deep learning model by using semantic.	Improve products, processes, and decisions.	Used a core manufacturing process ontology to design deep neural networks.	[83]
16	A methodology for the combination of ontology-based knowledge management and machine learning for the classification of multiple spatial data sources.	Nature conservation	Determination of the grassland indicators wetness and alkalinity.	Improve interoperability, reproducibility and exchange of data.	Used a decision tree classifier (DT) approach in combination with ontological formalism to generate classification procedures and results.	[84]
17	A roadmap is presented combining ontology and machine learning approaches for intelligent building monitoring.	Energy consumption	Intelligent monitoring of building energy consumption	Enhanced energy utilization.	Used nearest neighbour algorithm to predict the electricity consumption based on raw data along with knowledge data related to occupancy.	[85]

2. Ontology specification document for AquaONT (Table A2)

Table A 2. Ontology specification document for AquaONT describing the scope and purpose

Ontology Requirement Specification Document	
1	**Purpose**
	The purpose of AquaONT is to structure, model and store the aquaponics 4.0 system's knowledge, and use it to enable data-driven decision making for farmers by developing a functional decision support system.
2	**Domain**
	Agriculture – Smart Indoor Soilless Farming – Aquaponics 4.0
3	**Scope**
	The ontology has to focus on the aquaponics 4.0 system – which is a digital farm based on a smart farming concept. The level of granularity is directly linked with the indoor vertical farming terminologies.
4	**Implementation Language**
	The ontology has to be implemented in OWL 2 web ontology language.
5	**Intended End-Users**
	User 1. A farmer who wants to build a new indoor aquaponics 4.0 farm and wants to get information on the growth parameters of crops and fish. User 2. A person who wants to understand the impact of crops on the design configuration of aquaponics grow beds before building a physical farm for his/her startup. User 3. Researchers who want to build further applications such as decision support systems, and expert systems. User 4. Industrial practitioners who want to find out the causes of poor crop quality in their aquaponic farm.
6	**Intended Uses**
	Use 1. Develop a decision support system. The ontology model will be used to develop a system for crop quality monitoring and control. Use 2. Search for aquaponics 4.0 system information. A farmer will look for general information about the management of the aquaponics 4.0 system. Use 3. Search for optimal ranges of environmental and water parameters for the sustainability of aquaponics 4.0 farm. Use 4. Build an interface to enable parametric design automation. The ontology will be used to develop an application to visualize the impact of crop type on the design configuration of grow bed.
7	**Ontology Requirements**
	a. Non-Functional Requirements 1. The ontology must be based on the international and Canadian food standards in existence or under development. **b. Functional requirements** 1. Optimal ranges of environmental parameters: light intensity, humidity, temperature, etc. 2. Optimal ranges of water parameters: pH, DO, TDS, water level, flow rate, nitrogen level etc. 3. Types of crops: leafy green vegetables such as lettuce, basil, spinach, mint etc. 4. Optimal growth conditions for crop survival: environmental plus water parameters. 5. The growth cycle of each crop: seedling, sprouting, vegetative and plant. 6. Species of fish suitable for the crops considered: Tilapia, catfish, trout etc. 7. Optimal growth conditions for fish survival: environmental plus water parameters. 8. The growth cycle of each fish: fingerlings and mature fish.

	9. NFT grow channel configuration: standard widths, lengths, and heights available in the commercial market. 10. Specifications of fish tanks, mechanical filters, and biological filters. 11. Specifications of sensors and electronic devices installed at the farm for monitoring and controlling. 12. Crop and fish quality specifications and control standards determined by international and Canadian food agencies.

3. A formal model of AquaONT showing classes and subclasses (Figure A.1)

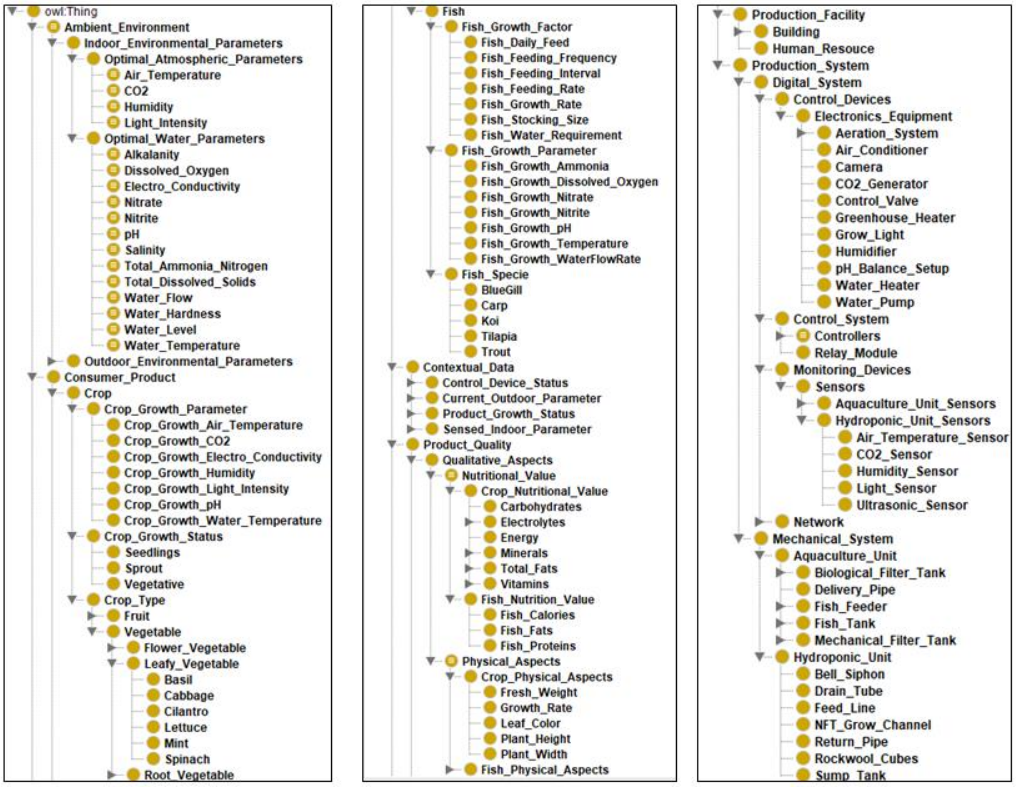

Fig A. 1. AquaONT formal model implemented on Protégé showing classes and subclasses

4. AquaONT instances for different classes and subclasses (Figure A.2)

Concept	Class	Subclass	Subclass	Instances
Contextual_Data	Sensed_Indoor_Parameters	AO:Sensed_Indoor_Parameter	AO:Sensed_pH	AO:6.7
		AO:Sensed_Indoor_Parameter	AO:Sensed_AT	AO:24°C
		AO:Sensed_Indoor_Parameter	AO:Sensed_WT	AO:25°C
		AO:Sensed_Indoor_Parameter	AO:Sensed_LDR	AO:478Lumens
		AO:Sensed_Indoor_Parameter	AO:Sensed_EC	AO:1000µS/cm
		AO:Sensed_Indoor_Parameter	AO:Sensed_RH	AO:80
Consumer_Product	Crop	AO:Crop_Growth_Parameter	AO:Crop_Growth_CO2	AO:Lettuce_Growth_CO2
		AO:Crop_Growth_Parameter	AO:Crop_Growth_Water_Temperature	AO:Lettuce_Growth_WaterTemperature
		AO:Crop_Growth_Parameter	AO:Crop_Growth_Electro_Conductivity	AO:Lettuce_Growth_Electro_Conductivity
		AO:Crop_Growth_Parameter	AO:Crop_Growth_pH	AO:Lettuce_Growth_pH
		AO:Crop_Growth_Parameter	AO:Crop_Growth_Humidity	AO:Lettuce_Growth_Humidity
		AO:Crop_Growth_Parameter	AO:Crop_Growth_Light_Intensity	AO:Lettuce_Growth_Light_Intensity
		AO:Crop_Growth_Parameter	AO:Crop_Growth_Air_Temperature	AO:Lettuce_Growth_AirTemperature
	Fish	AO:Fish_Growth_Parameter	AO:Fish_Growth_Dissolved_Oxygen	AO:Tilapia_Growth_DO
		AO:Fish_Growth_Parameter	AO:Fish_Growth_WaterFlowRate	AO:Tilapia_Growth_WFR
		AO:Fish_Growth_Parameter	AO:Fish_Growth_pH	AO:Tilapia_Growth_pH
		AO:Fish_Growth_Parameter	AO:Fish_Growth_Nitrate	AO:Tilapia_Growth_Nitrate
		AO:Fish_Growth_Parameter	AO:Fish_Growth_Temperature	AO:Tilapia_Growth_Temperature
		AO:Fish_Growth_Parameter	AO:Fish_Growth_Nitrite	AO:Tilapia_Growth_Nitrite
		AO:Fish_Growth_Parameter	AO:Fish_Growth_Ammonia	AO:Tilapia_Growth_Ammonia
Production_System	Mechanical_System	AO:Hydroponic_Unit	AO:NFT_Grow_Channel	AO:NFT_Grow_Channel_01
		AO:Hydroponic_Unit	AO:Sump_Tank	AO:Sump_Tank_01
		AO:Hydroponic_Unit	AO:Drain_Tube	AO:Drain_Tube_01
		AO:Hydroponic_Unit	AO:Return_Pipe	AO:Return_Pipe_01
	Digital_System	AO:Hydroponic_Unit_Sensors	AO:Humidity_Sensor	AO:Humidity_Sensor_01
		AO:Hydroponic_Unit_Sensors	AO:CO2_Sensor	AO:CO2_Sensor_01
		AO:Hydroponic_Unit_Sensors	AO:Ultrasonic_Sensor	AO:Ultrasonic_Sensor_01
		AO:Hydroponic_Unit_Sensors	AO:Air_Temperature_Sensor	AO:Air_Temperature_Sensor_01
Product_Quality	Crop_Nutritional_Value	AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Vitamin_C_Content
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Carbohydrates_Content
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Calcium_Content
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Vitamin_A_Content
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Total_Fats_Content
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Protein_Content
	Crop_Physical _Aspects	AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Fresh_Weight
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Leaf_Color
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Plant_Width
		AO:Leafy_Vegetable	AO:Lettuce	AO:Lettuce_Plant_Height

Fig A. 2. AquaONT instances for different classes and subclasses

5. Object properties of AquaONT (Figure A.3)

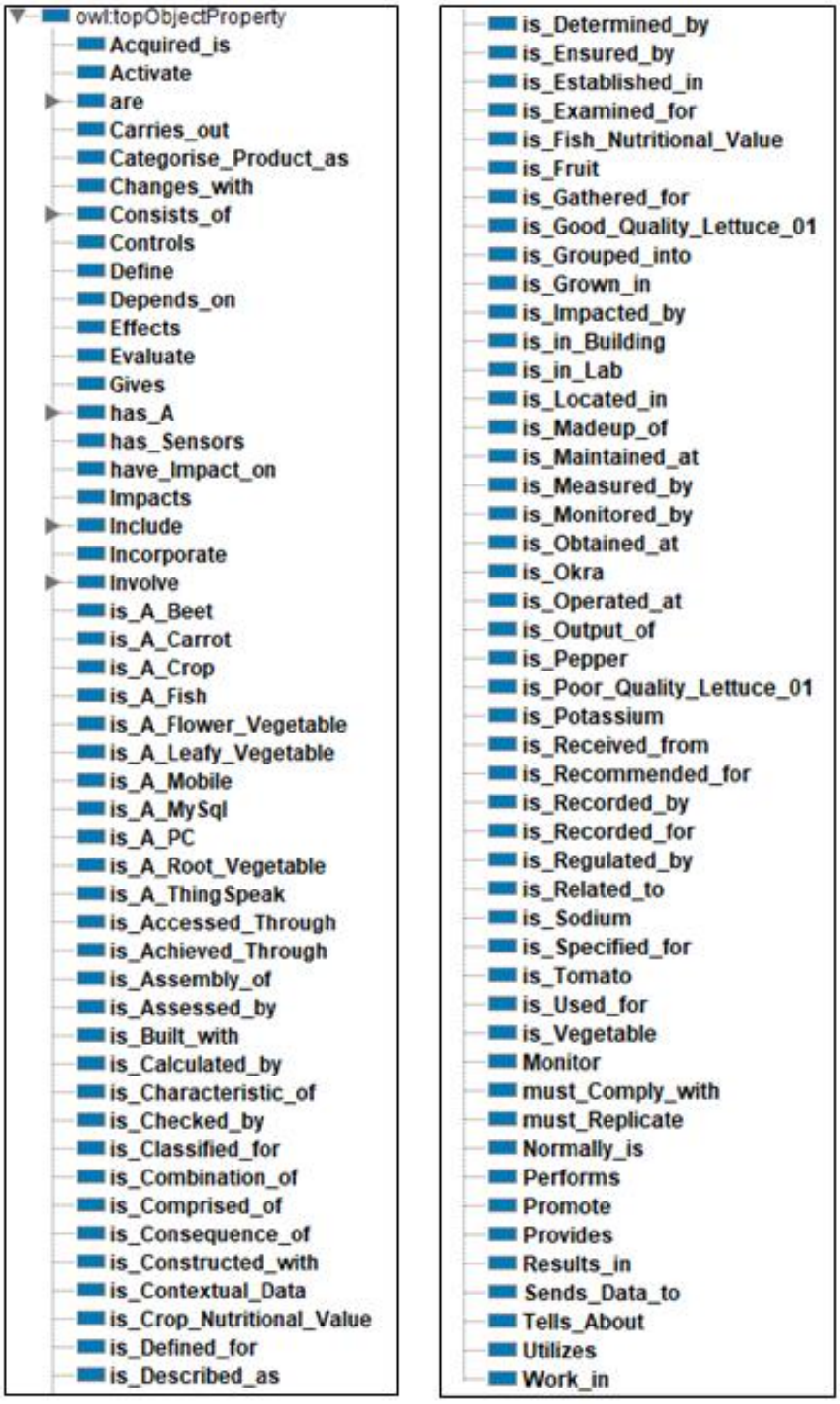

Fig A. 3. Object properties of AquaONT formal model

6. Object properties showing relationships between classes and subclasses (Figure A.4)

Domain	ObjectProperty	Range
AO:Contextual_Data	AO:Acquired_is	AO:Sensed_Indoor_Parameter
AO:Arduino	AO:Collects_Data_From	AO:Sensors
AO:Controllers	AO:Commonly_Used_are	AO:Arduino
AO:Sensors	AO:Generally_are	AO:Aquaculture_Unit_Sensors
AO:Sensors	AO:Generally_are	AO:Hydroponic_Unit_Sensors
AO:Humidity_Sensor	AO:Gives	AO:Sensed_RH
AO:Humidity	AO:Impacts	AO:Humidity_Sensor
AO:Fish_Growth_Factor	AO:Incorporate	AO:Fish_Daily_Feed
AO:Fish_Growth_Factor	AO:Incorporate	AO:Fish_Feeding_Frequency
AO:Fish_Growth_Factor	AO:Incorporate	AO:Fish_Feeding_Interval
AO:Control_System	AO:Controls	AO:Control_Devices
AO:Product_Inspection	AO:Define	AO:Nutritional_Inspection
AO:Product_Inspection	AO:Define	AO:Physical_Inspection
AO:Qualitative_Value	AO:Depends_on	AO:Indoor_Environmental_Parameters
AO:Crop_Growth_Parameter	AO:Normally_Include	AO:Crop_Growth_Air_Temperature
AO:Crop_Growth_Parameter	AO:Normally_Include	AO:Crop_Growth_CO2
AO:Crop_Growth_Parameter	AO:Normally_Include	AO:Crop_Growth_Electro_Conductivity
AO:Crop_Growth_Parameter	AO:Normally_Include	AO:Crop_Growth_Humidity
AO:Crop_Growth_Parameter	AO:Normally_Include	AO:Crop_Growth_Light_Intensit
AO:Optimal_Water_Parameters	AO:Mainly_Include	AO:Total_Dissolved_Solids
AO:Optimal_Water_Parameters	AO:Mainly_Include	AO:Water_Flow
AO:Optimal_Water_Parameters	AO:Mainly_Include	AO:Water_Hardness
AO:Optimal_Water_Parameters	AO:Mainly_Include	AO:Water_Level
AO:Optimal_Water_Parameters	AO:Mainly_Include	AO:Water_Temperature
AO:Optimal_Water_Parameters	AO:Mainly_Include	AO:pH
AO:Optimal_Atmospheric_Parameters	AO:Mainly_Involve	AO:Humidity
AO:Optimal_Atmospheric_Parameters	AO:Mainly_Involve	AO:Air_Temperature
AO:Optimal_Atmospheric_Parameters	AO:Mainly_Involve	AO:CO2
AO:Crop_Physical_Aspects	AO:are_Evaluated_by	AO:Fresh_Weight
AO:Crop_Physical_Aspects	AO:are_Evaluated_by	AO:Growth_Rate
AO:Crop_Physical_Aspects	AO:are_Evaluated_by	AO:Leaf_Color
AO:Crop_Physical_Aspects	AO:are_Evaluated_by	AO:Plant_Height
AO:Crop_Nutritional_Value	AO:is_Assessed_by	AO:Minerals
AO:Crop_Nutritional_Value	AO:is_Assessed_by	AO:Vitamins
AO:Crop_Nutritional_Value	AO:is_Assessed_by	AO:Carbohydrates
AO:Crop_Nutritional_Value	AO:is_Assessed_by	AO:Electrolytes
AO:Crop_Nutritional_Value	AO:is_Assessed_by	AO:Energy
AO:Crop_Nutritional_Value	AO:is_Assessed_by	AO:Protein
AO:Aquaculture_Unit_Sensors	AO:Considered_are	AO:Alkalinity_Sensor
AO:Aquaculture_Unit_Sensors	AO:Considered_are	AO:Dissolved_Oxygen_Sensor
AO:Aquaculture_Unit_Sensors	AO:Considered_are	AO:Electrical_Conductivity_Sensor
AO:Aquaculture_Unit_Sensors	AO:Considered_are	AO:Flow_Sensor
AO:Aquaculture_Unit_Sensors	AO:Considered_are	AO:Nitrite_Sensor
AO:Aquaculture_Unit_Sensors	AO:Considered_are	AO:Total_Ammonia_Nitrogen_Sensor
AO:Hydroponic_Unit	AO:is_Built_with	AO:Drain_Tube
AO:Hydroponic_Unit	AO:is_Built_with	AO:Feed_Line
AO:Hydroponic_Unit	AO:is_Built_with	AO:NFT_Grow_Channel
AO:Hydroponic_Unit	AO:is_Built_with	AO:Return_Pipe
AO:Hydroponic_Unit	AO:is_Built_with	AO:Rockwool_Cubes
AO:Hydroponic_Unit	AO:is_Built_with	AO:Sump_Tank
AO:Aquaculture_Unit	AO:is_Constructed_with	AO:Mechanical_Filter_Tank
AO:Aquaculture_Unit	AO:is_Constructed_with	AO:Biological_Filter_Tank
AO:Aquaculture_Unit	AO:is_Constructed_with	AO:Fish_Tank
AO:Aquaculture_Unit	AO:is_Constructed_with	AO:Delivery_Pipe
AO:Aquaculture_Unit	AO:is_Constructed_with	AO:Fish_Feeder

Fig A. 4. Object properties showing relationships between classes and subclasses

7. Datatype properties of AquaONT (Figure A.5)

Fig A. 5. Datatype properties of AquaONT

8. Datatype properties showing relationships between instances and attributes (Figure A.6)

Instance	DatatypeProperty	Attribute
AO:Lettuce_Growth_CO2	AO:hasGrowthValue	"1000 ppm"^^
AO:Lettuce_Growth_LightIntensity	AO:hasGrowthValue	"15-17 mols/m^2/day of PAR or PPFD 650-800nm wavelength"^^
AO:Optimal_Relative_Humidity	AO:hasLowerOptimalValue	"60.0"^^xsd:float
AO:Optimal_Relative_Humidity	AO:hasUpperOptimalValue	"80.0"^^xsd:float
AO:Light_Sensor_01	AO:hasModelNo	"SI1147-M01-GM"
AO:Light_Sensor_01	AO:hasaSupplier	"Digi Key"
AO:Light_Sensor_01	AO:hasaType	"Ambient, IR, Ultraviolet (UV)"
AO:Light_Sensor_01	AO:hasManufacturer	"Silicon Labs"
AO:Light_Sensor_01	AO:hasIDNo	"LS01"
AO:Light_Sensor_01	AO:hasSupplyVoltage	"1.71V ~ 3.6V"
AO:NFT_Grow_Channel_01	AO:isMadeupof	"PVC"
AO:NFT_Grow_Channel_01	AO:isUsedfor	"Lettuce Growing"
AO:NFT_Grow_Channel_01	AO:hasaSupplier	"Crop King"
AO:NFT_Grow_Channel_01	AO:hasCapacity	"135 Plants"
AO:NFT_Grow_Channel_01	AO:hasIDNo	"NGC01"
AO:NFT_Grow_Channel_01	AO:hasA	"1 inch Square Holes"

Fig A. 6. Datatype properties showing relationships between instances and attribute

References

[1] Al-Kodmany K. The vertical farm: A review of developments and implications for the vertical city. Buildings, 2018,8(2):24–60.
[2] Wei Y, Li W, An D, Li D, Jiao Y, Wei Q. Equipment and intelligent control system in aquaponics: A review. IEEE Access, 2019,7:169306–169326.
[3] Martinez P, Ahmad R. AllFactory : an aquaponics 4.0 transdisciplinary educational and applied research learning factory at the University of Alberta, In Proceedings of the 2021 Conference on Learning Factories; 2021. p.5–7.
[4] Rhayem A, Mhiri A, Gargouri F. Semantic web technologies for the Internet of Things: systematic literature review. Internet of Things, 2020,11:100206–10027.
[5] Aydin S, Aydin MN. Semantic and syntactic interoperability for agricultural open-data platforms in the context of IoT using crop-specific trait ontologies. Applied Sciences, 2020,10(13):4460–4486.
[6] Drury B, Fernandes R, Moura MF, de Andrade Lopes A. A survey of semantic web technology for agriculture. Information Processing in Agriculture, 2019,6(4):487–501.
[7] Munir K, Sheraz Anjum M. The use of ontologies for effective knowledge modeling and information retrieval. Applied Computing and Informatics, 2018,14(2):116–126.
[8] Martinez-Cruz C, Blanco IJ, Vila MA. Ontologies versus relational databases: Are they so different? A comparison. Artificial Intelligence Review, 2012,38(4):271–290.
[9] Abbasi R, Martinez P, Ahmad R. An ontology model to support the automated design of aquaponic grow beds. Procedia CIRP, 2021,100:55–60.
[10] Yanes AR, Martinez P, Ahmad R. Towards automated aquaponics: A review on monitoring, IoT, and smart systems. Journal of Cleaner Production, 2020,263:121571–121585.
[11] Tyson RV, Danyluk MD, Simonne EH, Treadwell DD. Sustainable aquaponic vegetable and fish co-production. In Proceedings of the Florida State Horticultural Society, 2012,125. p.381–385.
[12] Aceto G, Persico V, Pescapé A. A survey on information and communication technologies for Industry 4.0: State-of-the-art, taxonomies, perspectives, and challenges. IEEE Communications Surveys and Tutorials, 2019,21(4):3467–3501.
[13] Hari Kumar N, Baskaran S, Hariraj S, Krishnan V. An autonomous aquaponics system using 6LoWPAN based WSN. In Proceedings of 4th International Conference on Future Internet of Things and Cloud Workshops, IEEE, 2016. p.125–132.
[14] Hwang J, Yoe H. Study on the context-aware middleware for ubiquitous greenhouses using wireless sensor networks. Sensors, 2011,11(5):4539–4561.
[15] Davenport T, de Long D, Beers M. Successful Knowledge management projects. Sloan Management Review. link: https://sloanreview.mit.edu/article/successful-knowledge-management-projects/. 1998.
[16] Devedzic V. Knowledge modeling - State of the art. Integrated Computer-Aided Engineering, 2001,8(3):257–281.
[17] Bimba AT, Idris N, Al-Hunaiyyan A, Mahmud RB, Abdelaziz A, Khan S, Change V. Towards knowledge modeling and manipulation technologies: A survey. International Journal of Information Management, 2016,36(6):857–871.
[18] Gruber TR. A translation approach to portable ontology specifications. Knowledge Acquisition, 1993,5:1000–1006.
[19] Zhang F, Ma ZM, Li W. Storing OWL ontologies in object-oriented databases. Knowledge-Based Systems, 2015,76:240–255.
[20] Ramos L. Semantic web for manufacturing, trends and open issues: Toward a state of the art. Computers and Industrial Engineering, 2015, 90:444–460.
[21] Maalel A, Mejri L, Mabrouk HH, ben Ghezela H. Towards an ontology of help to the modeling of accident scenario: Application on railroad transport. In the Proceeding of 6th IEEE International Conference on Sciences of Electronics, Technologies of Information and Telecommunications, 2012. p.1–6.
[22] Zhang CX, Cao CG, Gu F, Si JX. Domain-specific formal ontology of archaeology and its application in knowledge acquisition and analysis. Journal of Computer Science and Technology, 2004,19(3): 290-301.
[23] Zhang C, Cao C, Sui Y, Wu X. A Chinese time ontology for the semantic web. Knowledge-Based Systems, 2011, 24(7):1057-1074.
[24] Miller E. An introduction to the Resource Description Framework. Bulletin of the American Society for Information Science and Technology, 1998, 25(1):15–19.
[25] World wide web consortium. W3C organisation. link: https://www.w3.org/OWL/. 2012.
[26] Allemang D, Hendler J. Semantic Web for the Working Ontologist: Effective modeling in RDFS and OWL. Elsevier. Link: http://www.kevenlw.name/downloads/Ontologist.pdf . 2011 /2020 (12:37 PM).
[27] Antoniou G, Groth P, Harmelen F van, Hoekstra R. A Semantic Web Primer (Third Edition). link: https://mitpress.mit.edu/books/semantic-web-primer-third-edition. 2012.
[28] Horrocks I, Patel-Schneider PF, Bechhofer S, Tsarkov D. OWL rules: A proposal and prototype implementation. Journal of Web Semantics, 2005,3(1):23–40.
[29] Musen MA. The protégé project: a look back and a look forward. AI Matters, 2015,1(4):4–12.
[30] Kim M, Kang H, Kwon S, Lee Y, Kim K, Pyo CS. Augmented ontology by handshaking with machine learning. In the Proceedings of International Conference on Advanced Communication Technology, 2017.p.740–743.
[31] Sheth A, Ramakrishnan C. Semantic (Web) technology in action: Ontology-driven information systems for search, integration and analysis. IEEE Data Engineering Bulletin, 2003,26(4):40–50.
[32] Hagedorn T, Bone M, Kruse B, Grosse I, Blackburn M. Knowledge representation with ontologies and semantic web technologies to promote augmented and artificial intelligence in systems engineering. Insight, 2020, 23(1).p.15–20.
[33] Petrenko SA, Makoveichuk KA. Ontology of cybersecurity of self-recovering smart Grid. In Proceedings of CEUR Workshop, 2017. p.98–106.

[34] Sharma N, Mangla M, Mohanty SN, Gupta D, Tiwari P, Shorfuzzaman M, Rawashdeh M. A smart ontology-based IoT framework for remote patient monitoring. Biomedical Signal Processing and Control, 2021,68:102717–102728.
[35] Chukkapalli SSL, Mittal S, Gupta M, Abdelsalam M, Joshi A, Sandhu R, Joshi K. Ontologies and artificial intelligence systems for the cooperative smart farming ecosystem. IEEE Access, 2020,8:45–64.
[36] Dutta S, Chukkapalli SSL, Sulgekar M, Krithivasan S, Das PK, Joshi A. Context-sensitive access control in smart home environments. In the Proceedings of IEEE 6th International Conference on Big Data Security on Cloud, 2020. p.35–41.
[37] AGROVOC, Food and Agriculture Organizations of the United Nations, link: https://agrovoc.uniroma2.it/agrovoc/agrovoc/en/. 2019.
[38] Arnaud E, Cooper L, Shrestha R, Menda N, Nelson RT, Matteis L, Skofic M, Bastow, Ruth, Jaiswal P, Mueller L, McLaren G. Towards a reference plant trait ontology for modeling knowledge of plant traits and phenotypes. In the Proceedings of 4th International Conference on Knowledge Engineering and Ontology Development, KEOD, 2012,2. p. 220–225.
[39] Thunkijjanukij A, Kawtrakul A, Panichsakpatana S, Veesommai U. Ontology development: A case study for thai rice. Agriculture and Natural Resources, 2009,43(3):594–604.
[40] Sodanil M, Phonarin P, Porrawatpreyakorn N. An ontology-based query expansion for an agricultural expert retrieval system. In the Proceedings of ACM International Conference, 2012. p.358–362.
[41] Phonarin P, Nitsuwat S, Haruechaiyasak C. AGRIX: An ontology-based agricultural expertise retrieval framework. Advanced Materials Research, 2012,403:3714–3718.
[42] Pokharel S, Sherif MA, Lehmann J. Ontology-based data access and integration for improving the effectiveness of farming in Nepal. In the Proceedings of IEEE International Joint Conference on Web Intelligence and Intelligent Agent Technology, 2014, 2. p.319–326.
[43] Afzal H, Kasi MK. Ontology-based knowledge modeling for rice crop production. In the Proceedings of International Conference on Future Internet of Things and Cloud, 2019. p.343–350.
[44] Abrahao E, Hirakawa AR. Task ontology modeling for technical knowledge representation in agriculture field operations domain. In the Proceedings of 2nd IEEE International Conference on Information Systems Engineering, 2017. p. 12–16.
[45] Aydin S, Aydin MN. Ontology-based data acquisition model development for agricultural open data platforms and implementation of OWL2MVC tool. Journal of Computers and Electronics in Agriculture, 2020,175:105589–105609.
[46] Sree S, Chukkapalli L. Ontology-driven AI and access control systems for smart fisheries. In the Proceedings of ACM Workshop on Secure and Trustworthy Cyber-Physical Systems, 2021. p.59–68.
[47] Sivamani S, Bae N, Cho Y. A smart service model based on ubiquitous sensor networks using vertical farm ontology. International Journal of Distributed Sensor Networks, 2013,9(12):161495–161502.
[48] Sivamani S, Kim H, Lee M, Park J, Shin C, Cho Y. An Ontology Model for Smart Service in Vertical Farms – An OWL-S Approach. International Journal of U and e-Service, Science and Technology, 2016,9(1):161–170.
[49] Kim T, Bae N, Lee M, Shin C, Cho Yongyun, Park J. A Study of an agricultural ontology model for an intelligent service in a vertical farm. International Journal of Smart Home, 2013,7(4):117–125.
[50] Husna N, Hasbullah M, Zakaria Z. The development of STIF in ontology based on the methontology approach. In the Proceedings of UTM Computing Innovation in Computing Technology and Applications, 2017,2. p.1–7.
[51] Fernandez M, Gómez-Pérez A, Juristo N. Methontology: from ontological art towards ontological engineering. In the Proceedings of the AAAI97 Spring Symposium Series on Ontological Engineering, 1997. p.33–40.
[52] Ahmad R, Masse C, Jituri S, Doucette J, Mertiny P. Alberta Learning Factory for training reconfigurable assembly process value stream mapping. Procedia Manufacturing, 2018,23:237–242.
[53] Scattini N, Maj SP. Aquaponics – A process control approach. Modern Applied Science, 2017,11(11):43–48.
[54] Somerville C, Cohen M, Pantanella E, Stankus A, Lovatelli A. Small-scale aquaponic food production. Integrated fish and plant farming. Link: http://www.fao.org/3/i4021e/i4021e.pdf . 2014 /2021 (1:17 PM).
[55] Zambri MKM, Aras MSM, Khamis A, Shah HNM, Harun MH, Yusof AA. Design and development of intelligent aquaponics system. International Journal of Engineering and Technology (UAE), 2018,7:410–424.
[56] Gu T, Wang XH, Pung HK, Zhang DQ. An ontology-based context model in intelligent environments. ArXiv. link: https://arxiv.org/abs/2003.05055/. 2020.
[57] Abbasi R, Reyes A, Martinez E, Ahmad R. Real-time implementation of a digital twin for robot-based production line. In Proceedings of the 2021 Conference on Learning Factories; 2021. p.1–2.
[58] Brosnan T, Sun DW. Improving quality inspection of food products by computer vision - A review. Journal of Food Engineering, 2004, 61(1):3–16.
[59] Kyaw TY, Ng AK. Smart Aquaponics System for Urban Farming. Energy Procedia, 2017, 143:342–347.
[60] Estellés A, Alcina A. A model for formalizing characteristics in Protégé-OWL. In the Proceedings of CEUR Workshop, 2009. p.1–6.
[61] Health Canada. Nutrient value of some common foods. Canadian nutrient file. Link: http://www.diabetesclinic.ca/en/diab/4nutrition/Nutrient_Content.pdf. 2005 /2021 (2:09 PM)
[62] Abburu S. A Survey on Ontology reasoners and comparison. International Journal of Computer Applications, 2012, 57(17):33–39.
[63] Noy NF, Fergerson RW, Musen MA. The knowledge model of protégé-2000: Combining interoperability and flexibility. In Proceedings of International Conference on Knowledge Engineering and Knowledge Management, Berlin, 2000. p.17–32.
[64] Zhang C, Zhou G, Lu Q. Decision support-oriented ontological modeling of product knowledge. In the Proceedings of IEEE 2nd Information Technology, Networking, Electronic and Automation Control Conference, 2017. p.39–43.
[65] Maboko MM, du Plooy CP. Effect of plant spacing on growth and yield of lettuce in a soilless production system. South African Journal of Plant and Soil, 2009,26(3):195–198.

[66] Sharma N, Acharya S, Kumar K, Singh N, Chaurasia OP. Hydroponics as an advanced technique for vegetable production: An overview. Journal of Soil and Water Conservation, 2018,17(4):364–371.
[67] Goddek S, Joyce A, Kotzen B, Burnell GM. Aquaponics food production systems: combined aquaculture and hydroponic production technologies for the future. Cham: Springer Nature, 2019. p.1–619.
[68] Varga M, Csukas B, Kucska B. Implementation of an easily reconfigurable dynamic simulator for recirculating aquaculture systems. Aquacultural Engineering, 2020, 90:102073–102085.
[69] Karimanzira D, Keesman KJ, Kloas W, Baganz D, Rauschenbach T. Dynamic modeling of the INAPRO aquaponic system. Aquacultural Engineering, 2016,75:29–45.
[70] Amin MT, Kissock JK. Dynamic modeling and verification of an energy-efficient greenhouse with aquaponics. In Proceedings of ASME 2016 10th International Conference on Energy Sustainability collocated with the ASME 2016 Power Conference and the ASME 2016 14th International Conference on Fuel Cell Science, Engineering and Technology, 2016,1. p.1–9.
[72] Survey A, Kushala VM, Supriya MC. Deep learning ontology : Dimensions in the field of agriculture, A survey. International Journal of Latest Technology in Engineering, Management & Applied Science, 2018,7(1):17–21.
[73] Naja I, Markovic M, Edwards P, Cottrill C. A semantic framework to support AI system accountability and audit. In Proceedings of European Semantic Web Conference, 2021. p.160–176.
[74] Ayadi A, Samet A, de Beuvron FDB, Zanni-Merk C. Ontology population with deep learning-based NLP: a case study on the Biomolecular Network Ontology. Procedia Computer Science, 2019,159:572–581.
[75] Raaijmakers S, Brewster C. Exploiting ontologies for deep learning: A case for sentiment mining. In the Proceedings of CEUR Workshop, 2018. p.1–4.
[76] Rahman H, Hussain MI. A lightweight dynamic ontology for the Internet of Things using machine learning technique. ICT Express, 2020,7(3):355-360.
[77] Umbrico A, Orlandini A, Cesta A. An Ontology for Human-Robot Collaboration. Procedia CIRP, 2020,93:1097–1102.
[78] Zheng H, Wang Y, Han C, Le F, He R, Lu J. Learning and applying ontology for machine learning in cyber-attack detection. In the Proceedings of 17th IEEE International Conference on Trust, Security and Privacy in Computing and Communications/12th IEEE International Conference on Big Data Science and Engineering, 2018. p.1309–1315.
[79] Swathi Lakshmi V, Nithya V, Sripriya K, Preethi C, Logeshwari K. Prediction of diabetes patient stage using ontology-based machine learning system. In the Proceedings of IEEE International Conference on System, Computation, Automation and Networking, 2019. p.2019–2022.
[80] Steinmetz C, Rettberg A, Ribeiro FGC, Schroeder G, Pereira CE. Internet of things ontology for digital twin in cyber-physical systems. In 8th IEEE Brazilian Symposium on Computing System Engineering, 2018. p.154–159.
[81] Rogushina J, Gladun A, Pryima S. Use of ontologies for metadata records analysis in big data. In the Proceeding of CEUR Workshop, 2018. p.46–63.
[82] Alkahtani M, Choudhary A, De A, Harding JA. A decision support system based on ontology and data mining to improve design using warranty data. Journal of Computers and Industrial Engineering, 2019,128:1027–1039.
[83] Santodomingo R, Rohjans S, Uslar M, Rodríguez-Mondéjar JA, Sanz-Bobi MA. Ontology matching system for future energy smart grids. Journal of Engineering Applications of Artificial Intelligence, 2014,32:242–257.
[84] Huang X, Zanni-Merk C, Crémilleux B. Enhancing deep learning with semantics: An application to manufacturing time series analysis. Procedia Computer Science, 2019,159:437–446.
[85] Moran N, Nieland S, Tintrup gen. Suntrup G, Kleinschmit B. Combining machine learning and ontological data handling for multi-source classification of nature conservation areas. International Journal of Applied Earth Observation and Geoinformation, 2017,54:124–133.
[86] Delgoshaei P, Heidarinejad M, Austin MA. Combined ontology-driven and machine learning approaches to monitoring of building energy consumption. In the Proceeding of Building Performance Modeling Conference and SimBuild, 2018. p. 667-674.